

Basic Structure of a Django Program

Django is a **Python web framework** used to develop dynamic and database-driven web applications.

A typical Django project has the following structure:

```
myproject/
|
├─ manage.py
|
├─ myproject/
|   ├─ __init__.py
|   ├─ settings.py
|   ├─ urls.py
|   ├─ asgi.py
|   └─ wsgi.py
|
└─ myapp/
    ├─ __init__.py
    ├─ admin.py
    ├─ apps.py
    ├─ models.py
    ├─ views.py
    ├─ tests.py
    └─ urls.py
```

1. manage.py

- A command-line utility used to manage the Django project.
- Used to run the development server, create applications, perform migrations, etc.

Eg:

python manage.py runserver

2. settings.py

Contains the **configuration settings** of the project.

Examples:

- Database configuration
- Installed applications
- Security settings
- Templates
- Static files

3. `urls.py`

Defines the **URL routing** of the application.

Example:

```
from django.urls import path
from myapp import views
```

```
urlpatterns = [
    path('home/', views.home),
]
```

4. `views.py`

Contains the **business logic** of the application.

Example:

```
from django.http import HttpResponse

def home(request):
    return HttpResponse("Welcome to Django")
```

5. `models.py`

Defines the **database structure** using Python classes.

Example:

```
from django.db import models

class Student(models.Model):
    name = models.CharField(max_length=100)
    age = models.IntegerField()
```

6. Templates

Templates are usually HTML files used to create the **user interface**.

Example:

```
<h1>Welcome to My Website</h1>
<p>Hello {{ name }}</p>
```

7. admin.py

Used to register models with Django's **built-in administration interface**.

```
from django.contrib import admin
from .models import Student
```

```
admin.site.register(Student)
```

8. wsgi.py and asgi.py

These files help Django communicate with web servers and are mainly used when **deploying** a Django application.